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Chapter 4

Series not on TV

4.1 Problem 4.1

Determine whether each of the following converges conditionally, converges absolutely, or diverges. You do not need to prove your answers, but state which of the following tests gives the answers: the k^{th} -term test, the geometric series test, and the alternating series test.

- | | | |
|---|---|--|
| (a) $\sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k}}$ | (d) $\sum_{k=1}^{\infty} \frac{1}{k!}$ | (g) $\sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt[3]{k}}$ |
| (b) $\sum_{k=1}^{\infty} (-1)^k \frac{k}{k+7}$ | (e) $\sum_{k=1}^{\infty} (\sqrt{k+1} - \sqrt{k})$ | (h) $\sum_{k=53}^{\infty} \frac{\ln(k)}{\ln(\ln(k))}$ |
| (c) $\sum_{k=1}^{\infty} \frac{1}{(\ln(4))^k}$ | (f) $\sum_{k=1}^{\infty} \frac{1}{2}$ | (i) $\sum_{k=7}^{\infty} \left(\frac{2}{k} + \frac{3}{k^2} \right)$ |

4.2 Problem 4.3

Prove that if $\sum_{k=1}^{\infty} |a_k|$ converges, then $\sum_{k=1}^{\infty} a_k$ converge too. This is Proposition 4.18.

4.3 Problem 4.5

Give an example of a series where $\sum_{k=1}^{\infty} a_k$ converges but $\sum_{k=1}^{\infty} a_{2k}$ diverges.

4.4 Problem 4.7

Give an example of a series $\sum_{k=1}^{\infty} a_k$ where:

- $\sum_{k=1}^{\infty} a_k$ converges,
- $\sum_{k=1}^{\infty} a_k^2$ diverges
- $\sum_{k=1}^{\infty} a_k^3$ converges

4.5 Problem 4.9

The geometric series test (Proposition 4.9) says that a geometric series diverges if $|r| \geq 1$. Recall that a series can diverge to ∞ , to $-\infty$, or can be “does not exist.” Which form of divergence is it for a geometric series? Note that your answer may depend on r .

4.6 Problem 4.11

- (a) Find a way to write $77.77777777\dots$ as a geometric series, and then prove this number is rational by using the geometric series test to write this number as a fraction with integers in the numerator and denominator.
- (b) Write $77.77777777\dots$ as a different geometric series, and use the geometric series test to write this number as a fraction with integers in the numerator and denominator. Are your two fractions the same?
- (c) A number q has a repeating decimal if the non-integer portion of its decimal expansion is repetitive. For example, $72.578578578578578\dots$ has a repeating decimal. Prove that if a number q has a repeating decimal, then q is rational.

4.7 Problem 4.13

- (a) Prove that if (ka_k) converges to a nonzero real number L , then the series $\sum_{k=1}^{\infty} a_k$ diverges. Give an example to show that the converse is false.
- (b) Prove that if (k^2a_k) converges (to any real number), then the series $\sum_{k=1}^{\infty} a_k$ converges. Give an example to show that the converse is false.

4.7.1 Solution (a)

Proof. We know that (ka_k) converges to a nonzero L .

$$\forall \varepsilon > 0, \exists N : \forall k > N, |ka_k - L| < \varepsilon \quad (4.1)$$

It is very much possible that $\varepsilon = \frac{|L|}{2}$. We can also apply the reverse triangle inequality and isolate $|a_k|$.

$$\frac{|L|}{2} = \varepsilon > |ka_k - L| \geq ||ka_k| - |L|| \quad (4.2)$$

$$\frac{|L|}{2} > |ka_k| - |L| > -\frac{|L|}{2} \quad (4.3)$$

$$\frac{3|L|}{2} > |ka_k| > \frac{|L|}{2} \quad (4.4)$$

$$\frac{3|L|}{2} > k|a_k| > \frac{|L|}{2} \quad (4.5)$$

$$\frac{3|L|}{2k} > |a_k| > \frac{|L|}{2k} = \frac{|L|}{2} \cdot \frac{1}{k} \quad (4.6)$$

Because of its convergence to L , beyond a certain point, a_k has the same sign as L . If we only work beyond that point, we can remove the absolute values. This gives us two cases. First is the case where $L > 0$.

$$\frac{3L}{2k} > a_k > \frac{L}{2} \cdot \frac{1}{k} \quad (4.7)$$

The other case is the one where $L < 0$.

$$-\frac{3L}{2k} > -a_k > -\frac{L}{2} \cdot \frac{1}{k} \quad (4.8)$$

We can substitute in a new sequence (b_k) in case 2 where for any k , $b_k = -a_k$ to get back to the point described in case 1. Since it's the same point, I won't be replacing a_k with b_k .

$$\frac{3L}{2k} > a_k > \frac{L}{2} \cdot \frac{1}{k} \quad (4.9)$$

Now, recall the harmonic series.

$$\sum_{k=1}^{\infty} \frac{1}{k} \rightarrow \infty \quad (4.10)$$

We can multiply this in both cases by $\frac{L}{2}$.

$$\sum_{k=1}^{\infty} \frac{L}{2} \cdot \frac{1}{k} \rightarrow \infty \quad (4.11)$$

Since the harmonic series diverges and (a_k) is always greater than our above alteration of the harmonic series, $\sum_{k=1}^{\infty} a_k$ diverges. OEΔ

For an example of how the converse could be false, consider the series $\sum_{k=1}^{\infty} k$. It diverges, and the components of it, even when multiplied by k , also diverge.

4.7.2 Solution (b)

Proof. We know that $(k^2 a_k)$ converges to a real number L .

$$\forall \varepsilon > 0, \exists N : \forall k > N, |k^2 a_k - L| < \varepsilon \quad (4.12)$$

We consider this based on whether L is or is not equal to zero.

Case 1: $L \neq 0$. If $L \neq 0$, it is very much possible that $\varepsilon = \frac{|L|}{2}$. We can

also apply the reverse triangle inequality and isolate $|a_k|$.

$$\frac{|L|}{2} = \varepsilon > |k^2 a_k - L| \geq ||k^2 a_k| - |L|| \quad (4.13)$$

$$\frac{|L|}{2} > |k^2 a_k| - |L| > -\frac{|L|}{2} \quad (4.14)$$

$$\frac{3|L|}{2} > |k^2 a_k| > \frac{|L|}{2} \quad (4.15)$$

$$\frac{3|L|}{2} > k^2 |a_k| > \frac{|L|}{2} \quad (4.16)$$

$$\frac{3|L|}{2k^2} = \frac{3|L|}{2} \cdot \frac{1}{k^2} > |a_k| > \frac{|L|}{2k^2} = \frac{|L|}{2} \cdot \frac{1}{k^2} \quad (4.17)$$

By the series p-test, we can recognize the convergence of $\sum_{k=1}^{\infty} \frac{1}{k^2}$, and multiply that by a constant.

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \rightarrow 0 \quad (4.18)$$

$$\sum_{k=1}^{\infty} \frac{3|L|}{2} \cdot \frac{1}{k^2} \rightarrow 0 \quad (4.19)$$

By the comparison test, this means that $\sum_{k=1}^{\infty} |a_k|$ converges. This in turn means that $\sum_{k=1}^{\infty} a_k$ converges.

Now for the other case.

Case 2: $L = 0$. We can rewrite the initial definition of convergence based on this.

$$\forall \varepsilon > 0, \exists N : \forall k > N, |k^2 a_k| < \varepsilon \quad (4.20)$$

Now, suppose that $\varepsilon = 1$. Solve for $|a_k|$.

$$|k^2 a_k| < 1 \quad (4.21)$$

$$|a_k| < \frac{1}{k^2} \quad (4.22)$$

As with earlier, this means the sum converges. OEΔ

For the example disproving the converse, consider $(\sum_{k=1}^{\infty} k^{-\frac{3}{2}})$. This converges. However, $(k^2 a_k)$ is equal to $(\sqrt{k})$, which diverges to infinity.

4.8 Problem 4.15

Prove the Cauchy condensation test. That is, suppose that (a_n) is a decreasing sequence for which $a_k \rightarrow 0$. Prove that $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges.

4.8.1 Solution

First, prove that if $\sum_{k=1}^{\infty} a_k$ converges, $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges.

Proof. Start with convergent $\sum_{k=1}^{\infty} a_k$ for decreasing $a_k \rightarrow 0$. We can recall the definition of the two.

$$\forall \varepsilon > 0, \exists N : \forall k > N, |a_k| < \varepsilon \quad (4.23)$$

$$a_{k+1} < a_k \quad (4.24)$$

$$\sum_{k=1}^{\infty} a_k \rightarrow L_1 \quad (4.25)$$

Here, we can try grouping terms in (a_k) . There is an inequality that is extrapolated from this being a decreasing sequence.

$$\forall n \in \mathbb{N}, 2^{n+1} - 1 \geq j \geq 2^n \implies a_{2^n} \geq a_j \geq a_{2^{n+1}-1} \quad (4.26)$$

$$\sum_{k=2^n}^{2^{n+1}-1} a_k \geq 2^n a_{2^{n+1}} \geq 0 \quad (4.27)$$

We can furthermore sum this over n .

$$\sum_{n=1}^{\infty} \sum_{k=2^n}^{2^{n+1}-1} a_k \geq \sum_{n=1}^{\infty} 2^n a_{2^{n+1}} \geq 0 \quad (4.28)$$

The former term is equal to the converging $\sum_{k=1}^{\infty} a_k$. Therefore, the latter term also converges. The latter term can be turned into $\sum_{k=1}^{\infty} 2^k a_{2^k}$ by adding a few terms.

$$\sum_{k=1}^{\infty} 2^k a_{2^k} = 2 * \sum_{n=1}^{\infty} 2^n a_{2^{n+1}} + 2 * a_2 = \sum_{n=1}^{\infty} 2^{n+1} a_{2^{n+1}} + 2 * a_2 \quad (4.29)$$

None of these operations (multiplying by 2 and adding $2 * a_2$) compromise convergence. Ergo, $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges. OEΔ

Next, prove that if $\sum_{k=1}^{\infty} 2^k a_{2^k}$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.

Proof. We begin with the earlier statement.

$$\forall n \in \mathbb{N}, 2^{n+1} - 1 \geq j \geq 2^n \implies a_{2^n} \geq a_j \geq a_{2^{n+1}-1} \quad (4.30)$$

$$2^n a_{2^n} \geq \sum_{k=2^n}^{2^{n+1}-1} a_k \geq 0 \quad (4.31)$$

Sum this over n .

$$\sum_{n=1}^{\infty} 2^n a_{2^n} \geq \sum_{n=1}^{\infty} \sum_{k=2^n}^{2^{n+1}-1} a_k \geq 0 \quad (4.32)$$

The latter term converges. As described earlier, it is equal to the sum over $\mathbb{N}$ of (a_k) .

$$\sum_{n=1}^{\infty} \sum_{k=2^n}^{2^{n+1}-1} a_k = \sum_{k=1}^{\infty} a_k \quad (4.33)$$

Therefore, $\sum_{k=1}^{\infty} a_k$ converges. OEΔ

4.9 Problem 4.17

Prove the ratio test via the following steps. Given a series $\sum_{k=1}^{\infty} a_k$ with $a_k \neq 0$, assume that

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| =: r < 1$$

We will prove that the series converges absolutely.

- (a) Let q be such that $r < q < 1$. Explain why there is some N such that $n \geq N$ implies that $|a_{n+1}| \leq |a_n| \cdot q$.
- (b) Explain why $\sum_{k=1}^{\infty} |a_N| \cdot q^k$ necessarily converges.
- (c) Finally, use part (b) to prove that $\sum_{k=1}^{\infty} |a_k|$ converges.

4.10 Problem 4.18

Prove the root test via the following steps. Given a series $\sum_{k=1}^{\infty} a_k$ where each $a_k \geq 0$, assume that the limit $\lim_{k \rightarrow \infty} (a_k)^{1/k}$ exists. Call this limit ρ . Then this series converges if $\rho < 1$ and diverges if $\rho > 1$. (The test is inconclusive if $\rho = 1$.)

- (a) Suppose $\rho < 1$. Let $\varepsilon = \frac{1-\rho}{2}$ and $\rho_1 = \rho + \varepsilon$. Prove that there is some N for which $(a_n)^{1/n} < \rho_1$ for all $n > N$.
- (b) Prove that $\sum_{k=N}^{\infty} a_k$ converges by comparing it to a geometric series. Then conclude that $\sum_{k=1}^{\infty} a_k$ also converges.
- (c) Suppose $\rho > 1$. Let $\varepsilon = \frac{\rho-1}{2}$ and $\rho_2 = \rho - \varepsilon$. Prove that there is some N for which $(a_n)^{1/n} > \rho_2$ for all $n > N$.
- (d) Prove that $\sum_{k=N}^{\infty} a_k$ diverges by comparing it to a geometric series. Then conclude that $\sum_{k=1}^{\infty} a_k$ also diverges.

4.10.1 Solution (a)

Begin with the limit from earlier and recall its convergence definition (traditional and Cauchy).

$$\lim_{k \rightarrow \infty} (a_k)^{1/k} = \rho < 1 \quad (4.34)$$

$$\forall \varepsilon > 0, \exists N : \forall n > N, |(a_n)^{1/n} - \rho| < \varepsilon \quad (4.35)$$

$$\forall \varepsilon > 0, \exists N : \forall m, n > N, |(a_m)^{1/m} - (a_n)^{1/n}| < \varepsilon \quad (4.36)$$

Expand the first absolute value, then isolate $(a_n)^{1/n}$.

$$-\varepsilon < (a_n)^{1/n} - \rho < \varepsilon \quad (4.37)$$

$$\rho - \varepsilon < (a_n)^{1/n} < \rho + \varepsilon \quad (4.38)$$

4.11 Problem 4.20

- (a) Give an example of two divergent series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$, such that $\sum_{k=1}^{\infty} a_k b_k$ converges.
- (b) Give an example of two convergent series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$, such that $\sum_{k=1}^{\infty} a_k b_k$ diverges.
- (c) Prove that if $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ converge absolutely, then $\sum_{k=1}^{\infty} a_k b_k$ converges absolutely.

4.11.1 Solution (a)

$$a_k = b_k = \frac{1}{k} \quad (4.39)$$

$$a_k b_k = \frac{1}{k^2} \quad (4.40)$$

4.11.2 Solution (b)

$$a_k = b_k = (-1)^{k+1} \frac{1}{\sqrt{k}} \quad (4.41)$$

$$a_k b_k = \frac{1}{k} \quad (4.42)$$

4.11.3 Solution (c)

Proof. Note that these two converge absolutely. This means that their absolute value sums also converge.

$$\sum_{k=1}^{\infty} |a_k| \rightarrow L_1 \quad (4.43)$$

$$\sum_{k=1}^{\infty} |b_k| \rightarrow L_2 \quad (4.44)$$

Since they converge, we can use the contrapositive of the k^{th} -term test to determine what they converge to.

$$a_k \rightarrow 0; |a_k| \rightarrow 0 \quad (4.45)$$

$$b_k \rightarrow 0; |b_k| \rightarrow 0 \quad (4.46)$$

Take the member absolute value functions and multiply them together.

$$|a_k| |b_k| = |a_k b_k| \rightarrow 0 \quad (4.47)$$

Since they converge to 0, there is a point after which $|a_k| < 1$ and $|b_k| < 1$.

$$\exists N \in \mathbb{N} : \forall k > N, |a_k| < 1 \quad (4.48)$$

$$\exists N \in \mathbb{N} : \forall k > N, |b_k| < 1 \quad (4.49)$$

We can multiply the last term of (4.48) by $|b_k|$ on both sides.

$$|b_k| > |a_k| |b_k| = |a_k b_k| \quad (4.50)$$

By the comparison test, this means that the sum of $(|a_k b_k|)$ over k converges. From that, we can conclude that the sum of $(a_k b_k)$ over k converges.

$$\sum_{k=1}^{\infty} |a_k b_k| \rightarrow L_3 \quad (4.51)$$

$$\sum_{k=1}^{\infty} a_k b_k \rightarrow L_4 \quad (4.52)$$

OEΔ

4.12 Problem 4.22

Consider the sum $\sum_{k=1}^{\infty} a_k$, and define s_n to be this series' n^{th} partial sum; that is, $s_n = \sum_{k=1}^n a_k$. The series $\sum_{k=1}^{\infty} a_k$ is called Cesaro summable if the following converges:

$$\lim_{n \rightarrow \infty} \frac{s_1 + s_2 + \cdots + s_n}{n}$$

- (a) Prove that if $\sum_{k=1}^{\infty} a_k$ converges, then this series is Cesaro summable.
- (b) Prove by example that if $\sum_{k=1}^{\infty} a_k$ is Cesaro summable this does not imply that $\sum_{k=1}^{\infty} a_k$ converges.

4.12.1 Solution (a)

Proof. Since $\sum_{k=1}^{\infty} a_k$ converges, its series of partial sums (a_n) also converges. Recall Problem 3.29 in chapter 3, where we proved that if $(a_n) \rightarrow a$, $b_n := \frac{a_1 + a_2 + \cdots + a_n}{n} \rightarrow a$. Suppose we apply that to (s_n) .

$$s_n \rightarrow s \tag{4.53}$$

$$\frac{s_1 + s_2 + \cdots + s_n}{n} \rightarrow s \tag{4.54}$$

Conclsively, this means that (s_n) converges.

OE Δ

4.12.2 Solution (b)

Take the case of $a_n = (-1)^n$. The sum diverges to a non-existant limit. However, dividing by the number of terms makes it converge to 0.5.

4.13 Problem 4.23

Show that if $\sum_{k=1}^{\infty} a_k$ is conditionally convergent, then there exists a rearrangement of this sum which diverges to ∞ .

4.14 Problem 4.24

Show that if $\sum_{k=1}^{\infty} a_k$ is conditionally convergent, then there exists a rearrangement of this sum whose limit does not exist.

4.15 Problem 4.25

Assume that $\sum_{k=1}^{\infty} a_k$ is conditionally convergent. Define the limit superior of a sequence (s_n) (Notation: “ $\limsup_{n \rightarrow \infty} s_n$ ”) to be

$$\limsup_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(\sup_{m \geq n} s_m \right)$$

And define the limit inferior of a sequence (s_n) (Notation: “ $\liminf_{n \rightarrow \infty} s_n$ ”) to be

$$\liminf_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(\inf_{m \geq n} s_m \right)$$

Prove that for any $\alpha, \beta \in \mathbb{R} \cup \{\pm\infty\}$ with $\alpha \leq \beta$, there is a rearrangement of $\sum_{k=1}^{\infty} a_k$ whose sequence of partial sums, (s_n) , have

$$\limsup_{n \rightarrow \infty} s_n = \beta \quad \text{and} \quad \liminf_{n \rightarrow \infty} s_n = \alpha$$

4.15.1 Solution

Proof. We know that, for a conditionally convergent series, there exists a rearrangement that converges to any $L \in \mathbb{R} \cup \pm\infty$. Suppose defined a new variable δ such that $\delta = \frac{|\alpha - \beta|}{4}$. We can separate a_n into a sequence of its non-negative and negative terms, called (p_n) and (n_n) respectively.

$$\sum_{k=1}^{\infty} p_k \rightarrow \infty \tag{4.55}$$

$$\sum_{k=1}^{\infty} n_k \rightarrow -\infty \tag{4.56}$$

We can order (p_k) and (n_k) such that both converge to 0.

$$\forall \varepsilon > 0, \exists N : \forall k > N, |p_k| < \varepsilon \tag{4.57}$$

$$\forall \varepsilon > 0, \exists N : \forall k > N, |n_k| < \varepsilon \tag{4.58}$$

There exists some P_0 and N_0 that we can sum up to the points P_0 and N_0 that $|p_{P_0}| < |\alpha - \beta|$ and $|n_{N_0}| < |\alpha - \beta|$.

$$\sum_{k=1}^{P_0} p_k + \sum_{k=1}^{N_0} n_k = c_0 \tag{4.59}$$

Beyond that point, there exists some P_1 that we can sum to that will be greater than $\beta - \delta$.

$$\exists P_1 \geq 0 : c_0 + \sum_{k=P_0+1}^{P_1} p_k = b_1 \geq \beta - \delta \quad (4.60)$$

From there, we can sum over (n_n) until the total is less than $\alpha + \delta$.

$$\exists N_1 \geq 0 : \alpha < b_1 + \sum_{k=N_0+1}^{N_1} n_k = c_1 \leq \alpha + \delta \quad (4.61)$$

We can continue this for any $m \geq 1$.

$$\forall m \in \mathbb{N}, \exists P_m \geq P_{m-1} : \beta > c_{m-1} + \sum_{k=P_{m-1}+1}^{P_m} p_k = b_m \geq b_{m-1} \quad (4.62)$$

$$\forall m \in \mathbb{N}, \exists N_m \geq N_{m-1} : c_{m-1} > b_m + \sum_{k=N_{m-1}+1}^{N_m} n_k = c_m \leq \alpha \quad (4.63)$$

Summing this up over all m , we end up with a sum that covers all of (a_n) and which has a limit inferior of α and limit superior of β . OE Δ

4.16 Problem 4.26

Prove that if $\sum_{k=1}^{\infty} a_k$ converges absolutely to L , then any rearrangement of this sum also converges to L .

4.16.1 Solution

4.17 Problem 4.27

Prove that if each $a_k \geq 0$ and $\sum_{k=1}^{\infty} a_k = \infty$, then any rearrangement of this sum also diverges to ∞ .

4.18 Problem 4.29

There is a neat theorem which, in a small way, relies on the fact that the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges. Here's the theorem: Every positive rational number can be written as a finite sum of distinct numbers of the form $1/n$. For example, the positive rational number $\frac{7243}{4140}$ can be written like this:

$$\frac{7243}{4140} = \frac{1}{1} + \frac{1}{2} + \frac{1}{5} + \frac{1}{21} + \frac{1}{527} + \frac{1}{3,054,492}$$

Now, take a moment to appreciate how magical this seems, because once you see the algorithm which produces the decomposition, it might lose some of its shine. Here's how I found the above decomposition of $\frac{7243}{4140}$ (which I basically chose at random):

$$\begin{aligned} \frac{7243}{4140} - 1 &= \frac{3103}{4140} \\ \frac{3103}{4140} - \frac{1}{2} &= \frac{1033}{4140} \\ \frac{1033}{4140} &< \frac{1}{3} \text{ and } \frac{1}{4} \\ \frac{1033}{4140} - \frac{1}{5} &= \frac{41}{828} \\ \frac{41}{828} &< \frac{1}{6}, \frac{1}{7}, \dots, \frac{1}{20} \\ \frac{41}{828} - \frac{1}{21} &= \frac{11}{5796} \\ \frac{11}{5796} &< \frac{1}{22}, \frac{1}{23}, \dots, \frac{1}{526} \\ \frac{11}{5796} - \frac{1}{527} &= \frac{1}{3,054,492} \end{aligned}$$

- (a) If the fraction that we start with is much bigger than 1, like $\frac{244,406,536}{7}$, why is this not an issue? Why will subtracting off 1, and then $1/2$, and then $1/3$, and so on, eventually catch up to it?
- (b) Prove that if this operation is performed on a fraction $\frac{p}{q}$ which is less than 1, then the numerator necessarily decreases (in the above example, the numerators decreased at every stage: 7243, 3103, 1033, 41, 11, 1). Note: The algorithm always subtracts off the biggest possible reciprocal. That is, if $\frac{p}{q} < 1$, then we subtract off $\frac{1}{n}$ where $\frac{1}{n} < \frac{p}{q} < \frac{1}{n-1}$.

(c) Explain why this proves the theorem.

4.18.1 Solution (a)

By subtracting reciprocals of natural numbers, we will be subtracting consecutive terms of the Harmonic Series. The Harmonic series diverges, so for the rational number $\frac{p}{q}$ there will be a point P at which $\sum_{k=1}^P \frac{1}{k} < \frac{p}{q} < \sum_{k=1}^{P+1} \frac{1}{k}$.

4.18.2 Solution (b)

Proof. On any given reciprocal subtracted, there is going to be an initial rational number $\frac{p_1}{q_1}$ which we subtract $\frac{1}{n}$ from to get the unreduced rational number $\frac{p_2}{q_2}$.

$$\frac{p_2}{q_2} = \frac{p_1}{q_1} - \frac{1}{n} = \frac{np_1 - q_1}{nq_1} \quad (4.64)$$

Since $\frac{p_2}{q_2}$ is unreduced, we can say that $q_2 = nq_1$.

$$\frac{p_2}{q_2} = \frac{np_1 - q_1}{nq_1} \quad (4.65)$$

$$p_2 = np_1 - q_1 \quad (4.66)$$

Furthermore, $\frac{p_1}{q_1}$ lies in between two reciprocals of natural numbers, which we can extrapolate on.

$$\frac{1}{n-1} > \frac{p_1}{q_1} \quad (4.67)$$

$$q_1 > np_1 - p_1 \quad (4.68)$$

$$p_1 > np_1 - q_1 \quad (4.69)$$

We can put that into our above equation for p_2 .

$$p_2 < p_1 \quad (4.70)$$

Even if the rational number is reduced, this will still be the case, because the new p_2 will be less than the initial p_2 . OEΔ

4.18.3 Solution (c)

This is because eventually, the rational number will be reduced to a point where $\frac{1}{n} < \frac{p}{q} < \frac{1}{n-1}$. After that point, the numerator will start reducing, and keep reducing until it hits 1.